Rank ≥ 6 for two torsion groups

Problèmes ouverts — A. Dujella · 2026

Problem 2.6 (Dujella, 2026). Are there infinitely many elliptic curves over $\mathbb{Q}$ with torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ (respectively $\mathbb{Z}/6\mathbb{Z}$) and rank $\geqslant 6$?

Remark. There are infinitely many curves with rank $\geqslant 5$ and torsion groups $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/6\mathbb{Z}$ (Eroshkin (2009)).